Project Retrospective

Conversational Memory Intelligence System

Deliverable	7 - Engineering Journal and Project Retrospective
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Written	2026-07-18
Covers	2026-07-14 through 2026-07-18 (M0-M6 build, D6 verification, D7 itself)

How this document was produced -- read this first

This retrospective was drafted by Claude Code from the project's real, contemporaneous records (.genesis/checkpoints/M0.md-M6.md, PLAN.md's progress log, and git history), at the author's explicit direction and for the author's review before submission -- not generated and submitted unreviewed. Per the handbook's Sec. 12 accountability rule, the author is responsible for every claim here regardless of which tool drafted it first. The 'what I would change' section in particular is a starting point grounded in the project's own evidence trail, offered for the author to confirm, edit, or replace with their own judgment -- not a substitute for it.

1. Timeline

Five real work sessions, reconstructed from git history and the checkpoints' own recorded dates -- see journal/2026-07-14.md through journal/2026-07-18.md for full detail.

Date	What happened
2026-07-14	D1-D4 authored pre-Genesis; hybrid-Postgres architecture chosen over vector-only and bi-temporal-graph alternatives; M0 built and verified; M1 begun
2026-07-15	M1 implementation state committed; a 2-day gap opens between AI verification and human sign-off
2026-07-16	M1 quiz-me gate answered by a human, milestone closed; M2 begun, a real indexing gap found in pre-flight
2026-07-17	M2 closed -- hybrid search + ranking; a real ranking bug found and fixed via independent verification, not the build's own tests
2026-07-18	M3, M4, M5, M6 built and closed; Deliverable 6 verification artifacts written after finding the required verification/ folder empty; Deliverable 7 (this document) begun

2. Assumptions That Changed, and the Evidence That Changed Them

The handbook asks specifically which assumptions changed and what evidence caused the change -- not just what shipped. Six real cases from this project:

Assumption (believed at the time)	Evidence it was wrong	What changed
M1: the secrets pattern set already covers every category threat_model.md names	L4 VERIFY found concrete misses: SSN, JWT, bare hex/mixed-case tokens, Stripe/GitHub prefixes	Added the missing patterns + regression tests; re-verified
M2: gating abstention on the single top-ranked candidate is sufficient	L4 VERIFY showed a distractor with maxed-out importance/recency could outrank and suppress a genuinely relevant answer	Gate abstention per-candidate on semantic score instead
M4: existing DB roles are enough for a nightly cross-tenant job	RLS has no tenant-registry table for a sweep to enumerate through; reusing the cmis superuser was already ruled out at M0	Added a new, narrowly-scoped cmis_job role (3 grants, BYPASSRLS, never exposed over HTTP)
M5: a raw INSERT is an acceptable stand-in for the write path in a benchmark	It silently skipped M3's contradiction resolution -- the exact mechanism the benchmark's S2 metric measures	Switched the benchmark to drive the real write path (write_gate.pipeline.process_turn)
M6: write_gate/judge.py's docstring claim ('every decision logged since M1') is accurate	Checked directly: only a reason string was logged on rejection, never the candidate text	Added write_gate_decision table logging both kept and dropped decisions with real text
D6/D7: rigorous build-time verification work counts as 'done' for the deliverable	The handbook grades a standalone verification/ folder; it was completely empty despite real L4 VERIFY work existing inside .genesis/checkpoints/	Wrote the required artifacts from fresh, independently re-run evidence

3. Mistakes and Recoveries

Five real mistakes, and how each was actually caught -- not all by the same mechanism.

Mistake	Caught by
M0: SET app.tenant_id = %s isn't valid SQL (SET doesn't take bind params)	The tests themselves, on the very first run
M2: abstention gated on the top-by-final-score candidate, not per-candidate	L4 VERIFY (independent model) -- the build's own green test suite did not catch this
M5: benchmark's first attempt used a raw INSERT, silently skipping contradiction resolution	Noticed during the build itself, before L4 VERIFY ever ran
M6: judge.py's docstring claim about decision logging was false	A direct check during G0 pre-flight, before any prototype code was written
D6: verification/ was entirely empty despite substantial real underlying work	A deliberate gap analysis against the handbook's own artifact list, not assumed fine

What this list says, taken together

Only one of these five was caught by the author's own primary tests. The rest needed either a second, independent reviewer (L4 VERIFY), a deliberate re-check of a claim before building on top of it (M6), or a direct comparison against an external specification rather than the project's own internal sense of 'done' (D6). That pattern -- same-author tests sharing the same blind spots as the code they test -- is probably the single most transferable lesson from this project, and it is why the handbook requires independent verification rather than accepting the implementation agent's own assessment.

4. What I Would Design Differently in a Second Iteration

- Write journal entries live, per session, not reconstructed at the end. This entire journal and retrospective had to be reconstructed on 2026-07-18 from checkpoint records rather than written as the reasoning happened -- exactly what the handbook warns against, and the clearest process gap in this project.
- Treat verification/ as a living folder updated at each milestone close-out, not a one-time write-up at the end. The real verification work (L4 VERIFY) was rigorous throughout the build; it simply never got surfaced into the artifact shape the handbook actually grades until this was noticed during Deliverable 6. Updating test_plan.md/security_report.md incrementally would have caught this months earlier than a final gap analysis.
- Tighten the gap between an AI verifier's APPROVE and the human's own sign-off. M1 sat two days between the two; M5 and M6's quiz-me questions were answered by the assistant at the human's direction rather than independently. Neither is disqualifying on its own -- both are disclosed rather than hidden -- but a second iteration should close that loop same-day as a default, reserving delegation for a deliberate, occasional choice rather than the norm.
- Re-tune or replace the fixed RELEVANCE_FLOOR constant. It is the one metric (S6, abstention failure rate) that improved but did not reach the same zero-gate the other hard invariants hit. A second iteration would treat this as unfinished, not merely 'improved,' and consider an adaptive threshold instead of one hand-picked constant reused everywhere.
- Validate the write-gate classifier prototype (M6/P2) against real production judge decisions before claiming ADOPT, not only a rule-based stand-in. The 1.000 held-out agreement is evidence the training mechanism works, not evidence the real judge's fuzzier boundary is this cleanly learnable -- ADR-001's own stated graduation trigger is real decision volume, which this experiment does not yet have.

5. Material AI Assistance (Handbook Sec. 12)

Per the handbook's accountability rule, this section records where AI assistance was material and which decisions stayed with the author.

What AI (Claude Code) did

- Implemented the majority of the code across M0-M6 inside a loop-based workflow (L1 BUILD / L2 DEBUG / L3 RESEARCH / L4 VERIFY), each milestone gated by an independent verification pass.
- Drafted Deliverable 6's verification/ artifacts from fresh, independently re-run test/benchmark/security evidence.
- Drafted this journal and retrospective (Deliverable 7) from the project's real recorded history, explicitly for the author's review rather than as a finished, self-certified submission.

What stayed with the author

- Choosing among the three architecture alternatives at project kickoff (hybrid Postgres over vector-only or a bi-temporal graph).
- Resolving the M6 spec ambiguity in PLAN.md directly (which reading of 'P2' to build) rather than letting the AI pick either silently.
- Directing the assistant to answer the M5/M6 quiz-me questions, an explicit delegation choice logged with that provenance rather than presented as independent verification.
- Reviewing, correcting, and taking ownership of this retrospective and the journal entries before submission -- this draft is not the final word on 'what I would do differently'; it is a starting point built from the evidence trail, not a replacement for the author's own reflection.